

Design Write-Up

RAG AI Assistant — "Attention Is All You Need" Telegram Bot (n8n + Pinecone + Gemini)

Architecture Overview

The assistant runs as two n8n workflows. Ingestion downloads the source PDF, chunks it into ~400-token pieces with 50-token overlap, and inserts each chunk into Pinecone with document name, page, and chunk index as metadata, using Gemini for embeddings. The query workflow is triggered by Telegram and handed to an AI Agent with a Gemini chat model, per-chat conversation memory, and a vector-store search tool on the same index. A strict system prompt forces the Agent to always search first, answer only from retrieved context, and return an exact fallback sentence when nothing relevant is found. Both success and error outputs converge on the same Telegram-reply and Sheets-logging nodes, so every interaction is always answered and logged.

Design Decisions

Native n8n nodes (Pinecone Vector Store, Embeddings Google Gemini, Gemini Chat Model) are used instead of hand-built HTTP calls for a cleaner, more maintainable workflow. Chunking happens in code before reaching the vector store node, since n8n's default loader silently re-splits already-chunked text, which would have destroyed the 400-token sizing and discarded chunk-level metadata. Two distinct failure messages are used rather than one generic error: a genuine "no relevant content" result returns the required fallback sentence, while an actual Gemini/service failure is routed through a separate error output with a clearly different message. Gemini's raw output is also sanitized of markdown/LaTeX characters before sending, since Telegram's parser crashes on unmatched symbols.

Challenges Faced

The biggest issue was a silent double-chunking bug: the vector store's default loader re-split already-chunked input, turning 36 intended chunks into 66 smaller, inconsistent ones. This was caught by comparing item counts before and after the loader, and fixed with a custom, deliberately oversized text splitter that never triggers. A related issue saw chunk metadata silently reset to a generic format after a node was reconfigured, traced by comparing live output to the expected configuration. On retrieval, a conceptually correct question sometimes triggered the fallback message simply because its conversational wording didn't overlap with the paper's academic vocabulary — a known embedding-search limitation, mitigated by increasing topK and prompting the Agent to retry with more technical phrasing before giving up.